9.6 - Solving Motion Problems Using Parametric & Vector-Valued Functions

Learning Objective FUN-8.B: Determine values for positions and rates of change in problems involving planar motion.

We spend a great deal of time in Calculus AB discussing straight line or rectilinear motion ("an object travels along the x-axis and has a position of $s(t)=t^2-4t+5$, etc.."). We can now discuss **motion along curved paths**.

As an object moves along a curve in the plane, the coordinates x and y of its center of mass are each functions of time t. Rather than use f and g to represent two functions, we write $x=x(t)$ and $y=y(t)$. So, our position vector $\mathbf{r}(t)$ is in the form $\mathbf{r}(t)=x(t)\mathbf{i}+y(t)\mathbf{j}$.

The nice thing about doing this is that it is very easy to use derivatives of the vector-valued function **r** to find the object's velocity and acceleration. Remember that velocity and acceleration are both vector quantities having magnitude and direction.

DEFINITION: Velocity and Acceleration

If $x(t)$ and $y(t)$ are twice differentiable functions of t and **r** is a vector-valued function given by $\mathbf{r}(t)=x(t)\mathbf{i}+y(t)\mathbf{j}$ the velocity vector and acceleration vector at time t are

Velocity:

$$\mathbf{v}(t)=\mathbf{r}'(t)=x'(t)\mathbf{i}+y'(t)\mathbf{j}$$

Acceleration:

$$\mathbf{a}(t)=\mathbf{r}''(t)=x''(t)\mathbf{i}+y''(t)\mathbf{j}$$

Speed:

$$\|\mathbf{v}(t)\|=\|\mathbf{r}'(t)\|=\sqrt{[x'(t)]^2+[y'(t)]^2}$$

<u>Example 1</u>: Velocity, Acceleration and Speed

A particle moves along a plane curve described by $\mathbf{r}(t)=3\sin\left(\frac{t}{2}\right)\mathbf{i}+3\cos\left(\frac{t}{2}\right)\mathbf{j}$ $=\langle 3\sin\left(\frac{t}{2}\right), 3\cos\left(\frac{t}{2}\right)\rangle$

a.) Generate a rectangular equation that describes the particle's motion.

$\cos^2\left(\frac{t}{2}\right)+\sin^2\left(\frac{t}{2}\right)=1$

$9\cos^2\left(\frac{t}{2}\right)+9\sin^2\left(\frac{t}{2}\right)=9$

$x^2+y^2=9$

$\tan\theta=\sin\theta\cdot\sec\theta$

$\cot\theta=\cos\theta\cdot\csc\theta$

b.) Find the velocity vector, acceleration vector and speed. Interpret the speed.

$\vec{v}(t)=\langle \frac{3}{2}\cos\left(\frac{t}{2}\right), -\frac{3}{2}\sin\left(\frac{t}{2}\right)\rangle$

$\vec{a}(t)=\langle -\frac{3}{4}\sin\left(\frac{t}{2}\right), -\frac{3}{4}\cos\left(\frac{t}{2}\right)\rangle$

$\|\vec{v}(t)\|=\sqrt{\frac{9}{4}}=\frac{3}{2}$

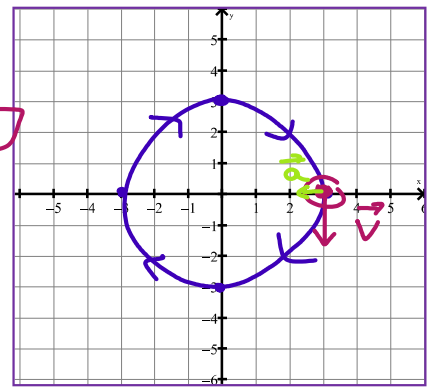
$\sqrt{\frac{9}{4}\cos^2\left(\frac{t}{2}\right)+\frac{9}{4}\sin^2\left(\frac{t}{2}\right)}$

Example 1: - cont'd

A particle moves along a plane curve described by $\mathbf{r}(t) = 3\sin\left(\frac{t}{2}\right)\mathbf{i} + 3\cos\left(\frac{t}{2}\right)\mathbf{j}$.

c.) Show the particle's path on the graph to the right. Be sure to show arrows to indicate direction.

t	x	y
0	0	3
π	3	0
2π	0	-3
3π	-3	0
4π	0	3



d.) Find the velocity and acceleration at $t = \pi$. Draw the velocity and acceleration vectors on the graph above.

$\vec{v}(\pi) = \langle 0, -\frac{3}{2} \rangle$

$\vec{a}(\pi) = \langle -\frac{3}{4}, 0 \rangle$

Scan the QR Code above to watch a video covering Example 1 (a and b)

Scan the QR Code above to watch a video covering Example 1 (c and d)

Example 2: Motion Along a Curve - I

Sketch the path of an object moving along the plane curve given by $\mathbf{r}(t) = (t^2 - 3)\mathbf{i} + t\mathbf{j}$ and find the velocity and acceleration vectors and speed at $t = 0$ and $t = 2$. Sketch the velocity and acceleration vectors on your curve below.

$\vec{r}(t) = \langle t^2 - 3, t \rangle$

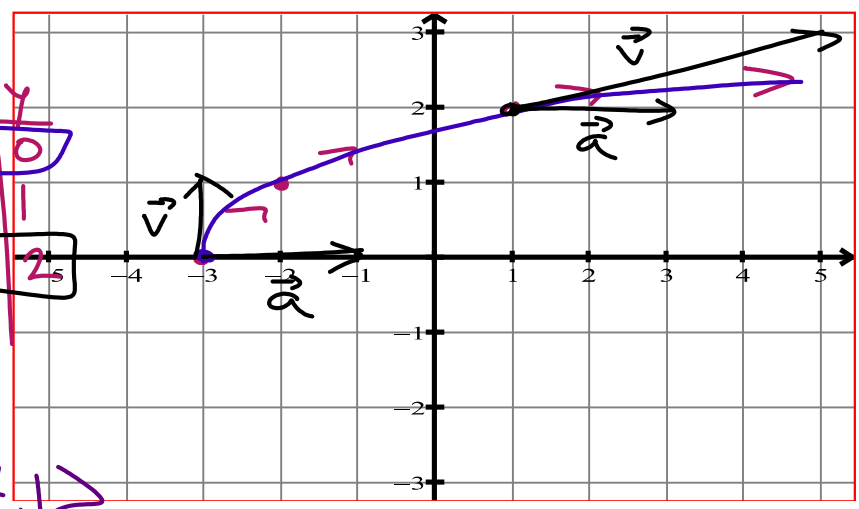
$\vec{v}(t) = \langle 2t, 1 \rangle$

$\vec{a}(t) = \langle 2, 0 \rangle$

$\|\vec{v}(t)\| = \sqrt{(2t)^2 + 1^2}$

$= \sqrt{4t^2 + 1}$

t	x	y
0	-3	0
1	-2	1
2	1	2



$\vec{v}(0) = \langle 0, 1 \rangle$

$\vec{a}(0) = \langle 2, 0 \rangle$

$\|\vec{v}(0)\| = 1$

$\vec{v}(2) = \langle 4, 1 \rangle$

$\vec{a}(2) = \langle 2, 0 \rangle$

$\|\vec{v}(2)\| = \sqrt{17}$

Scan the QR Code above to watch a video covering Example 2

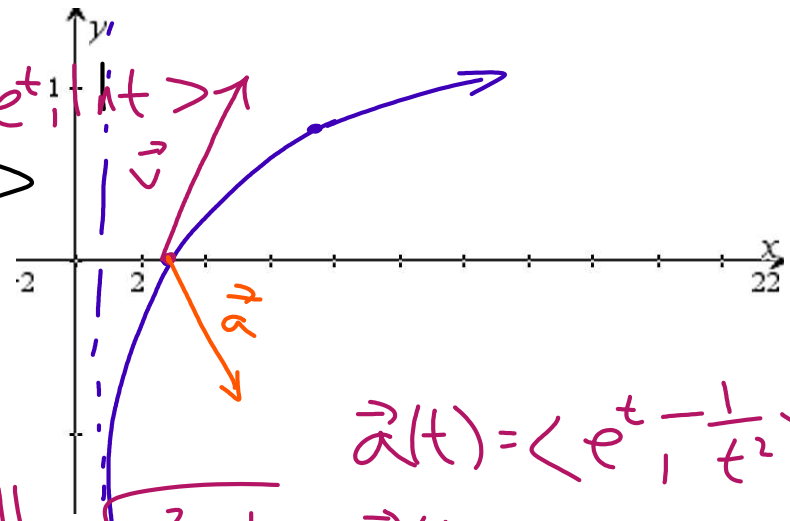
Example 3: Motion Along a Curve - II

a.) Sketch the path of an object moving along the plane curve given by $\mathbf{r}(t) = e^t\mathbf{i} + \ln t\,\mathbf{j}$. $= \langle e^t, \ln t \rangle$

$t > 0$

$\lim_{t \to 0^+} (\vec{r}(t)) = \langle 1, -\infty \rangle$

t	x	y
1	e	0
2	e^2	0.7



b.) Find the velocity and acceleration vectors and speed at $t = 1$.

$\vec{v}(t) = \langle e^t, t^{-1} \rangle$

$\vec{v}(1) = \langle e, 1 \rangle$

$\|\vec{v}(1)\| = \sqrt{e^2+1}$

$\vec{a}(t) = \langle e^t, -\frac{1}{t^2} \rangle$

$\vec{a}(1) = \langle e, -1 \rangle$

c.) Sketch the velocity and acceleration vectors on your curve above and to the right.

~~**d.)** Find the speed at $t = 1$.~~

e.) What happens to the position, velocity, and acceleration as t approaches infinity?

$\lim_{t \to \infty} \vec{r}(t) = \langle \lim_{t \to \infty} e^t, \lim_{t \to \infty} \ln t \rangle$

$\to \langle \infty, \infty \rangle$

up & right

$\lim_{t \to \infty} \vec{a}(t) = \langle \infty, 0 \rangle$

$\lim_{t \to \infty} \vec{v}(t) = \langle \lim_{t \to \infty} e^t, \lim_{t \to \infty} \frac{1}{t} \rangle = \langle \infty, 0 \rangle$

Scan the QR Code above to watch a video covering Example 3

Example 4: Motion Along a Curve - III

A particle starts at rest from the point $(-1,2)$ with an acceleration vector of $\mathbf{a}(t) = \mathbf{i} + 2\mathbf{j}$. Find the location of the particle at $t = 2$.

$\vec{a}(t) = \langle 1, 2 \rangle$

$\vec{r}(0) = \langle -1, 2 \rangle$

$\vec{v}(0) = \langle 0, 0 \rangle$

$\vec{v}(t) = \langle \int 1\,dt, \int 2\,dt \rangle$

$= \langle t + C_1, 2t + C_2 \rangle$

$= \langle t, 2t \rangle$

$C_1 = C_2 = 0, \; C_3 = -1, \; C_4 = 2$

$\vec{r}(t) = \langle \int t\,dt, \int 2t\,dt \rangle$

$= \langle \frac{t^2}{2} + C_3, t^2 + C_4 \rangle = \boxed{\langle \frac{t^2}{2} - 1, t^2 + 2 \rangle}$

Scan the QR Code above to watch a video covering Example 4

Example 5: Motion Along a Curve - IV

A particle with an initial velocity $\mathbf{i}+\mathbf{j}$ and initial position (1, 1) has acceleration vector $\mathbf{a}(t)=-\cos t\mathbf{i}+\sin t\mathbf{j}$ Find the location of the particle at $t=\pi$.

$\vec{r}(0)=\langle 1,1\rangle$

$\vec{v}(0)=\langle 1,1\rangle$

$C_1=1,\ C_2=2$

$\boxed{\vec{r}(\pi)=\langle \pi-1, 2\pi+1\rangle}$

$\vec{a}(t)=\langle -\cos t, \sin t\rangle$

$\vec{v}(t)=\langle \int -\cos t\,dt, \int \sin t\,dt\rangle$

$\vec{v}(t)=\langle -\sin t+C_1, -\cos t+C_2\rangle$

$\therefore\ \vec{v}(t)=\langle -\sin t+1, -\cos t+2\rangle$

$\vec{v}(0)=\langle C_1, C_2-1\rangle$

$\vec{r}(t)=\langle \cos t+t+C_3, -\sin t+2t+C_4\rangle$

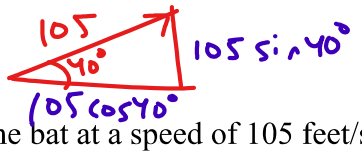
$\vec{r}(0)=\langle 1+C_3, C_4\rangle$

$C_3=0,\ C_4=1$

$\vec{r}(t)=\langle \cos t+t, -\sin t+2t+1\rangle$

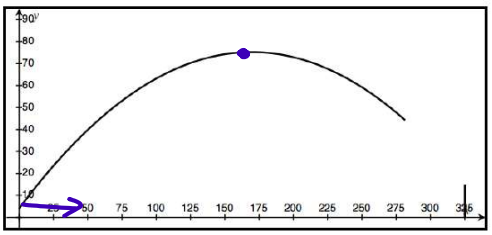
Scan the QR Code above to watch a video covering Example 5

Remember that when an object is subjected to gravity, its position s above the ground is given by $s=-16t^2+v_0t+s_0$. With that in mind, we can solve some vector motion problems that you may have done in the previous unit using parametric equations.



Example 6: Application of Motion- I

A baseball is hit 4 feet above the ground and leaves the bat at a speed of 105 feet/second at an angle of 40° to the ground.



a.) Write a vector that describes the motion of the ball.

$\langle 105\cos(40^\circ)\cdot t, -16t^2+105\sin(40^\circ)\cdot t+4\rangle$

b.) Determine how high the ball goes.

$\frac{d}{dt}\left(-16t^2+105\cdot\sin(40^\circ)\cdot t+4\right)=0$

$-32t+105\cdot\sin(40^\circ)=0$

$t=\frac{105\sin(40^\circ)}{32}=2.109\text{ s}$

$y(2.109)=75.176\text{ ft}$

c.) There is a 12-foot high fence 325 feet from home plate. Is the ball a home run?

$x(t)=325=105\cos(40^\circ)\cdot t$

$\Rightarrow t=\frac{325}{105\cos(40^\circ)}=4.041\text{ s}$

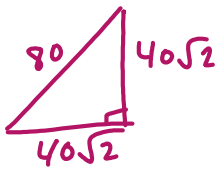
$y(4.041)=15.491\geq 12$

Home Run!

Scan the QR Code above to watch a video covering Example 6

Example 7: Application of Motion- II

Water from a fire hose is discharged at a maximum rate of 80 feet per second from a height 4 feet from the ground. Firefighters can get 25 feet from a burning building. They shoot the water from their hose at an angle of 45° to the ground.



$x(t) = 40\sqrt{2}\cdot t$

$y(t) = -16t^2 + 40\sqrt{2}\cdot t + 4$

a.) What is the maximum height that the water can reach?

$y'(t) = -32t + 40\sqrt{2}$

$y'(t) = 0 \rightarrow t = \frac{40\sqrt{2}}{32} = 1.768\ s$

$y(1.768) = 54\ ft$

b.) If the building completely burns down and the firemen are shooting water on the wreckage, how far into the wreckage can the water reach?

$y(t) = 0$

$\Rightarrow t = \frac{-40\sqrt{2} \pm \sqrt{(40\sqrt{2})^2 - 4(-16)(4)}}{-32}$

$\Rightarrow t = \frac{5\sqrt{2} + 3\sqrt{6}}{4} \approx 3.605\ s$

$x(3.605) = 178.923\ ft$

Scan the QR Code above to watch a video covering Example 7

Example 8: Application of Motion- III

A projectile is fired from the ground level at an angle of 10° the horizontal. Find the minimum velocity necessary if the projectile is to have a range of 100 feet.

$x(t) = v_0 \cos(10°)\cdot t$

$y(t) = -16t^2 + v_0 \sin(10°)\cdot t$

Let τ = time the projectile hits the ground

$y(\tau) = 0 \Rightarrow$

$\tau = \frac{v_0}{16}\sin(10°)$

Scan the QR Code above to watch a video covering Example 8

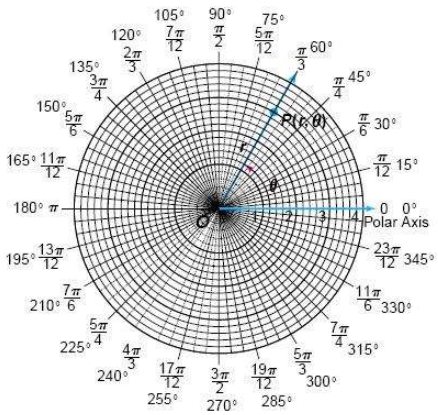
$x(\tau) = \frac{v_0^2}{16}\sin(10°)\cos(10°)$

$x(\tau) = 100 \Rightarrow v_0 = \frac{40}{\sqrt{\sin(10°)\cos(10°)}} \approx 96.727\ ft/s$

9.7 - Defining Polar Coordinates & Differentiating in Polar Form

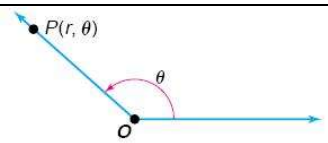
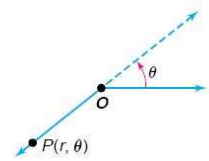
Learning Objective FUN-3.G: Calculate derivatives of functions written in polar coordinates.

Polar Coordinates



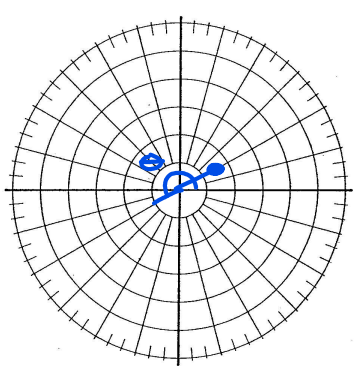
In a polar coordinate system, a fixed point O is called the **pole** or origin. The **polar axis** is usually a horizontal ray directed toward the right from the pole. The location of a point P in the polar coordinate system can be identified by polar coordinates in the form (r,θ). If a ray is drawn from the pole through P, the distance from the pole to point P is $|r|$. The measure of the angle formed by $\overline{OP}$ and the polar axis is θ. The angle can be measured in degrees or radians.

It is important to consider both positive and negative values of r.

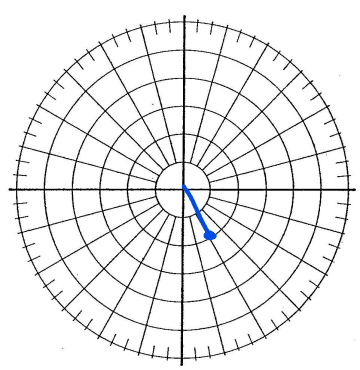
Suppose $r>0$. Then θ is the measure of any angle in standard position that has $\overline{OP}$ as its terminal side.	Suppose $r<0$. Then θ is the measure of any angle that has the ray opposite $\overline{OP}$ as its terminal side.
	

<u>Example 1</u>: Graph each point.

a. $P\left(-1.5,\frac{7\pi}{6}\right)$



b. $Q\left(2,-\frac{\pi}{3}\right)$



SCAN ME

Scan the QR Code above to watch a video covering Example 1

Coordinate Conversions

Converting Polar Coordinates to Rectangular Coordinates	The rectangular coordinates (x, y) of a point named by the polar coordinates (r, θ) can be found by using the following formulas. $x = r\cos\theta$ $y = r\sin\theta$

Converting Rectangular Coordinates to Polar Coordinates	The polar coordinates (r, θ) of a point named by the rectangular coordinates (x, y) can be found by the following formulas. $r = \sqrt{x^2 + y^2}$ $\theta = \text{Arctan}\,\frac{y}{x}$, when $x > 0$ $\theta = \text{Arctan}\,\frac{y}{x} + \pi$, when $x < 0$

Example 2: Convert the following polar points to rectangular points.

a.) $\left(2\sqrt{3}, \frac{\pi}{6}\right)$

$r \quad \theta$

$x = r\cos\theta = 2\sqrt{3}\cos\frac{\pi}{6} = 2\sqrt{3}\cdot\frac{\sqrt{3}}{2} = 3$

$y = r\sin\theta$

$y = 2\sqrt{3}\sin\frac{\pi}{6} = 2\sqrt{3}\cdot\frac{1}{2} = \sqrt{3}$

$(3, \sqrt{3})$

b.) $\left(8, \frac{5\pi}{4}\right)$

$x = 8\cos\left(\frac{5\pi}{4}\right) = 8\left(-\frac{\sqrt{2}}{2}\right) = -4\sqrt{2}$

$y = 8\sin\left(\frac{5\pi}{4}\right) = -4\sqrt{2}$

$(-4\sqrt{2}, -4\sqrt{2})$

SCAN ME

Scan the QR Code above to watch a video covering Example 2

Example 3: Convert the following rectangular points to polar points.

a.) $(3, -3)$ $\left(3\sqrt{2}, -\frac{\pi}{4}\right)$

$r = \sqrt{(3)^2 + (-3)^2} = \sqrt{18} = 3\sqrt{2}$

$\theta = \tan^{-1}\left(\frac{-3}{3}\right) = \tan^{-1}(-1) = -\frac{\pi}{4}$

b.) $(-4, 4\sqrt{3})$ $\left(8, \frac{2\pi}{3}\right)$

$r = \sqrt{(-4)^2 + (4\sqrt{3})^2} = 8$

$\theta = \tan^{-1}\left(\frac{4\sqrt{3}}{-4}\right) + \pi$

$= \tan^{-1}(-\sqrt{3}) + \pi = -\frac{\pi}{3} + \pi = \frac{2\pi}{3}$

SCAN ME

Scan the QR Code above to watch a video covering Example 3

Equation Conversions

Since points can be written in both rectangular and polar format, we can also write equations with both formats. Graphs that are linear are best written in rectangular format, while graphs that have curves are best written in polar form. Non-functions with loops are very difficult to write in rectangular form and usually need to be written as a piecewise function while in many cases, are very easily written in polar form. To convert from one form to another, we use the same relationships in the boxes above.

Example 4: Convert the following rectangular equations to polar equations. Verify by graphing both on your TI-Nspire using two different Graph pages.

a.) $y = 6$

$r\sin\theta = 6$

$r = \frac{6}{\sin\theta}$

$r = 6\csc\theta$

b.) $4x - 2y - 1 = 0$

$4r\cos\theta - 2r\sin\theta - 1 = 0$

$r = \frac{1}{4\cos\theta - 2\sin\theta}$

SCAN ME

Scan the QR Code above to watch a video covering Example 4

c. $y = 4x^2$

$r\sin\theta = 4r^2\cos^2\theta$

$\sin\theta = 4r\cos^2\theta$, $r \neq 0$

$r = \frac{\sin\theta}{4\cos^2\theta}$

$r = \frac{1}{4}\sec\theta\tan\theta$

Example 5: Convert the following polar equations to rectangular equations. Verify by graphing both on your TI-Nspire using two different Graph pages.

a.) $r = -\csc\theta = -\frac{1}{\sin\theta}$

$\sqrt{x^2+y^2} = -\frac{\sqrt{x^2+y^2}}{y}$

$1 = \frac{-1}{y}$

$y = -1$

$r\sin\theta = -1$

$y = -1$

b.) $r = \frac{2}{\sin\theta + 3\cos\theta}$

$r\sin\theta + 3r\cos\theta = 2$

$y + 3x = 2$

$y = -3x + 2$

c.) $r = \frac{3}{1 + 2\cos\theta}$

$r + 2r\cos\theta = 3$

$\sqrt{x^2+y^2} + 2x = 3$

$\sqrt{x^2+y^2} = 3 - 2x$

$x^2 + y^2 = 9 - 12x + 4x^2$

$y^2 = 9 - 12x + 3x^2$

$3x^2 - 12x - y^2 = 9$

SCAN ME

Scan the QR Code above to watch a video covering Example 5

To convert a graph in polar form to parametric form, we use the fact that $x = r\cos\theta$ and $y = r\sin\theta$.

Example 6: Convert the following polar equations to parametric equations.

a.) $r = 4$

$\sqrt{x^2+y^2} = 4$

$x^2 + y^2 = 16$

b.) $r = 4\sin\theta$

$r^2 = 4r\sin\theta$

$r^2 = 4y$

$\sqrt{x^2+y^2} = \frac{4y}{\sqrt{x^2+y^2}}$

$x^2 + y^2 = 4y$

$x^2 + y^2 - 4y + 4 = 4$

$x^2 + (y-2)^2 = 2^2$

c.) $r = \frac{4}{\cos\theta}$

$r\cos\theta = 4$

$x = 4$

SCAN ME

Scan the QR Code above to watch a video covering Example 6

Polar Graphs

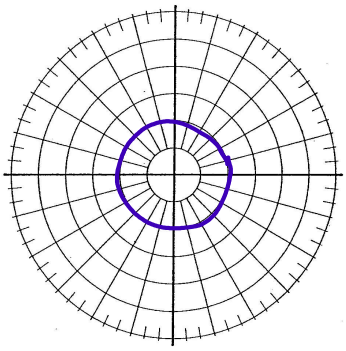
Example 7: Graphing Polar Equations

Graph each polar equation on the given polar coordinate planes.

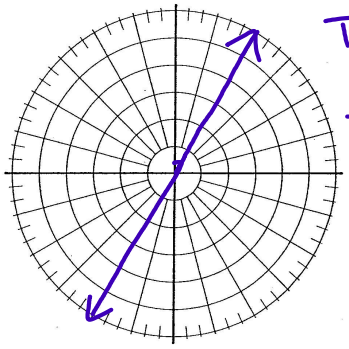
$x = 1$

$r\cos\theta = 1$

a.) $r = 2$

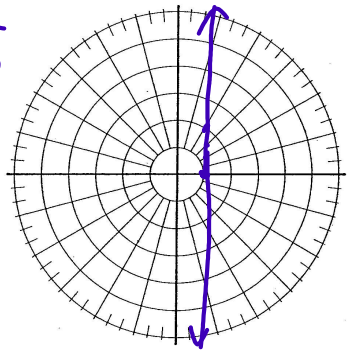


b.) $\theta = \frac{\pi}{3}$



θ	r
0	1
$\pi/6$	1.15
$\pi/4$	1.41
$\pi/3$	2

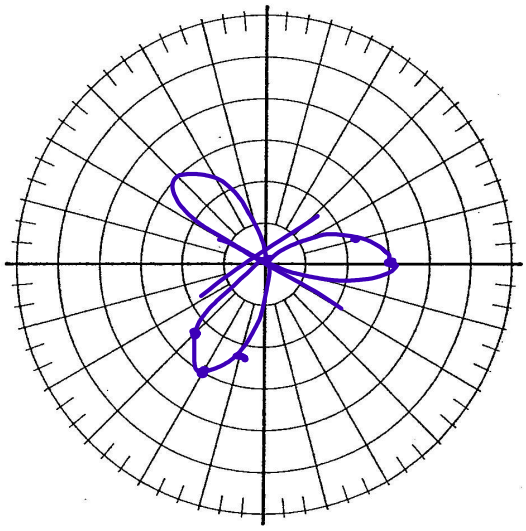
c.) $r = \sec\theta$



Example 8: Graphing Polar Equations

Sketch the graph of the polar equation $r = 3\cos 3\theta$ on the given polar coordinate plane.

θ	r
0	3
$\pi/12$	$\frac{3\sqrt{2}}{2} = 2.12$
$\pi/6$	0
$\pi/4$	$-\frac{3\sqrt{2}}{2}$
$\pi/3$	-3
$5\pi/12$	$-\frac{3\sqrt{2}}{2}$



SCAN ME

Scan the QR Code above to watch a video covering Example 7

SCAN ME

Scan the QR Code above to watch a video covering Example 8

Special Polar Graphs

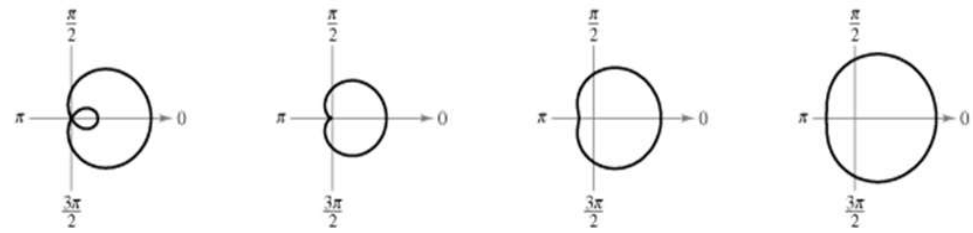
Limaçons

Limaçons

$r = a \pm b\cos\theta$

$r = a \pm b\sin\theta$

$(a > 0, b > 0)$



$\frac{a}{b} < 1$	$\frac{a}{b} = 1$	$1 < \frac{a}{b} < 2$	$\frac{a}{b} \geq 2$
Limaçon with inner loop	Cardioid (heart-shaped)	Dimpled limaçon	Convex limaçon

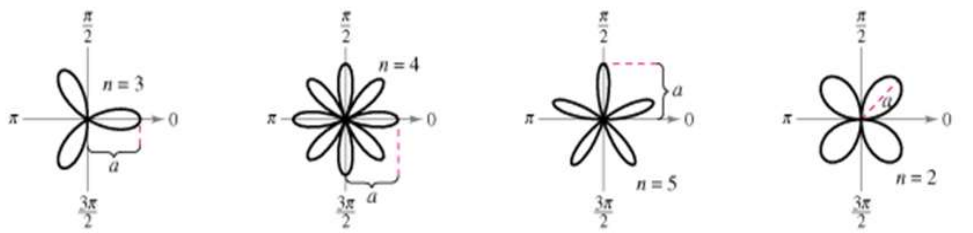
Rose Curves

Rose Curves

n petals if n is odd

$2n$ petals if n is even

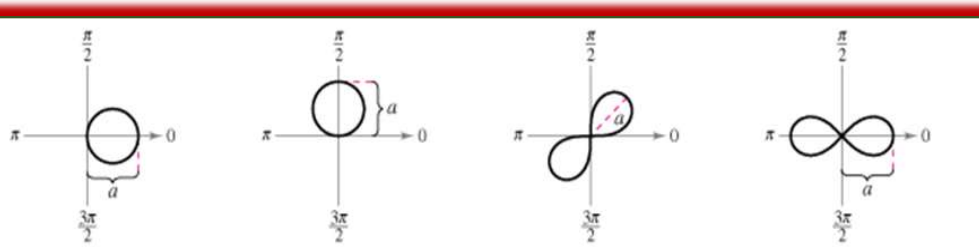
$(n \geq 2)$



$r = a\cos n\theta$	$r = a\cos n\theta$	$r = a\sin n\theta$	$r = a\sin n\theta$
Rose curve	Rose curve	Rose curve	Rose curve

Circles and Lemniscates

Circles and Lemniscates



$r = a\cos\theta$	$r = a\sin\theta$	$r^2 = a^2\sin 2\theta$	$r^2 = a^2\cos 2\theta$
Circle	Circle	Lemniscate	Lemniscate

Favorite Polar Graphs

Try graphing some of the following equations with your graphing calculator to see some very interesting polar graphs.

$r = \sin^3(2.5\theta) + \cos^3(2.5\theta)$ $\qquad$ $r = 2 + \sin^3(2.4\theta)$ $\qquad$ $r = \sin\left(\frac{8\theta}{5}\right)$

Slope and Tangent Lines

To find the slope of a tangent line to a polar graph, consider a differentiable function defined by $r = f(\theta)$. To find the slope in polar form, use the parametric equations

$x = r\cos\theta = f(\theta)\cos\theta$ and $y = r\sin\theta = f(\theta)\sin\theta$

Using the parametric form of dy/dx given in the previous section, you have

$$\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta}$$
$$= \frac{f(\theta)\cos\theta + f'(\theta)\sin\theta}{-f(\theta)\sin\theta + f'(\theta)\cos\theta}$$

THEOREM 9.7 SLOPE IN POLAR FORM

If f is a differentiable function of Θ, then the *slope* of the tangent line to the graph of $r = f(\theta)$ at the point (r, θ) is

$$\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{f(\theta)\cos\theta + f'(\theta)\sin\theta}{-f(\theta)\sin\theta + f'(\theta)\cos\theta}$$

Using the formula above, we make these similar observations (as we did with parametric equations):

- To find horizontal tangents to polar graphs, set $\frac{dy}{d\theta} = 0$ provided that $\frac{dx}{d\theta} \neq 0$.
- To find vertical tangents to polar graphs, set $\frac{dx}{d\theta} = 0$ provided that $\frac{dy}{d\theta} \neq 0$.
- If $\frac{dx}{d\theta}$ and $\frac{dy}{d\theta}$ are simultaneously zero, we can make no conclusions about tangent lines.

Example 9: Heart Examination

Consider the cardioid described by the polar equation $r = 1 + \sin\theta$.

a.) Find the slope of the tangent line to the cardioid when $\theta = \frac{\pi}{3}$.

$$\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta}$$

$y = (1+\sin\theta)\sin\theta$

$y = \sin\theta + \sin^2\theta$

$\frac{dy}{d\theta} = \cos\theta + 2\sin\theta\cos\theta$

$x = (1+\sin\theta)\cos\theta$

$x = \cos\theta + \sin\theta\cos\theta$

$\frac{dx}{d\theta} = -\sin\theta + \cos^2\theta - \sin^2\theta$

$= -\sin\theta + 1 - 2\sin^2\theta$

$$\frac{dy}{dx} = \frac{\cos\theta + 2\sin\theta\cos\theta}{-\sin\theta + \cos^2\theta - \sin^2\theta}$$

$\sin\theta = -\frac{1}{2}$

$$\left.\frac{dy}{dx}\right|_{\theta=\frac{\pi}{3}} = \frac{\frac{1}{2} + \cancel{2\cdot\frac{\sqrt{3}}{2}\cdot\frac{1}{2}}}{-\frac{\sqrt{3}}{2} + \frac{1}{4} - \frac{3}{4}} \cdot \frac{2}{2} = \frac{1+\sqrt{3}}{-\sqrt{3}-1} = \frac{1+\sqrt{3}}{-(1+\sqrt{3})} = -1$$

b.) Find the values for θ for which the cardioid's tangent lines are horizontal or vertical.

HTL $\Rightarrow \frac{dy}{d\theta} = 0$

$\cos\theta(1+2\sin\theta) = 0$

$\cos\theta = 0$

$\theta = \frac{\pi}{2}, \cancel{\frac{3\pi}{2}}, \frac{7\pi}{6}, \frac{11\pi}{6}$

VTL $\Rightarrow \frac{dx}{d\theta} = 0$

Example 10: Particle Motion of a Polar Curve

A polar curve r is given by $r = \theta + \cos\theta$, where $0 \le \theta \le 2\pi$. A particle is traveling along the polar curve r so that its position at time t is $(x(t), y(t))$ and such that $\frac{d\theta}{dt} = \frac{1}{2}$. Find $\frac{dx}{dt}$ at the instant $\theta = \frac{5\pi}{6}$, and interpret the meaning of your answer in the context of the problem.

$x = r\cos\theta$

$x = \theta\cos\theta + \cos^2\theta$

$$\frac{dx}{dt} = \left(\cos\theta - \theta\sin\theta - 2\sin\theta\cos\theta\right)\cdot\frac{d\theta}{dt}$$

$$\left.\frac{dx}{dt}\right|_{\theta=\frac{5\pi}{6}} = \left(-\frac{\sqrt{3}}{2} - \frac{5\pi}{6}\cdot\frac{1}{2} + \cancel{2\cdot\frac{1}{2}\cdot\frac{\sqrt{3}}{2}}\right)\cdot\frac{1}{2}$$

$$= -\frac{5\pi}{24} < 0$$

Moving to the left at $\theta = \frac{5\pi}{6}$

SCAN ME

Scan the QR Code above to watch a video covering Example 9

SCAN ME

Scan the QR Code above to watch a video covering Example 10